API Documentation: Shipments Module

Base URL: {{baseUrl}}/api/shipments

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1. Overview

The Shipments module handles the lifecycle of cargo movement, including manual creation, bulk CSV imports, status tracking, and Role-Based Access Control (RBAC).

2. Authentication & Authorization

- **Auth Type:** Bearer Token (JWT).
- **Header:** Authorization: Bearer <your_jwt_token>
- **Roles:** * ADMIN, OPERATIONS: Full Access (Create, Read, Update, Delete).
    - VIEWER: Read-only access to assigned shipments.

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3. Endpoint Specifications

3.1 Create Shipment (Manual)

Adds a single shipment record.

- **Method:** POST
- **Path:** /
- **Roles:** ADMIN, OPERATIONS
- **Request Body (JSON):**

JSON

```
{
  "shipment_id": "SHIP-8821",
  "client_name": "Global Traders Inc",
  "origin": "Port of Singapore",
  "destination": "London Gateway",
  "dispatch_date": "2023-11-01",
  "expected_delivery_date": "2023-11-20",
  "carrier_name": "Maersk"
}
```

- **Postman Success Response (201 Created):**

JSON

{
  "message": "Shipment created successfully",
  "id": "uuid-string-here"
}

3.2 Bulk Upload Shipments (CSV)

Uploads multiple shipments via a CSV file.

- **Method:** POST
- **Path:** /upload
- **Consumes:** multipart/form-data
- **Body:** * file: (Attach shipments.csv)
- **Postman Success Response (201/207):**

JSON

{
  "total_processed": 10,
  "successful_count": 8,
  "successes": ["SHIP-001", "SHIP-002"],
  "error_count": 2,
  "errors": [
    { "row": 5, "shipment_id": "SHIP-ERR", "issues": ["Invalid Date"] }
  ]
}

3.3 Get All Shipments

Retrieves a paginated list of shipments with filters.

- **Method:** GET
- **Path:** /
- **Query Parameters:**
    - page: (default 1)
    - limit: (default 50)
    - status: (Filter by enum, e.g., In Transit)

    - client: (Partial string match)
- **Postman Success Response (200 OK):**

JSON


```
{
  "total": 120,
  "page": 1,
  "limit": 50,
  "data": [
    {
      "shipment_id": "SHIP-8821",
      "status": "Created",
      "client_name": "Global Traders Inc",
      "created_at": "2023-10-31T12:00:00Z"
    }
  ]
}
```

3.4 Update Shipment Status

Updates the status, delivery date, or Proof of Delivery (POD).

- **Method:** PUT
- **Path:** /:id/status
- **Request Body (JSON):**

JSON


```
{
  "status": "Delivered",
  "delivered_date": "2023-11-18",
  "pod_received": true
}
```


- **Postman Success Response (200 OK):**

JSON


```
{
```

"id": "uuid-string",
"status": "Delivered",
"last_status_update": "2023-11-18T09:00:00Z"
}

3.5 Cancel Shipment (Soft Delete)

Marks a shipment as Cancelled.

- **Method:** DELETE
- **Path:** /:id
- **Postman Success Response (200 OK):**

JSON

{
"id": "uuid-string",
"status": "Cancelled"
}

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4. Postman Testing Steps

1. **Environment Setup:** Create a Postman Environment and set baseUrl.
2. **Token Retrieval:** Run your Login request. Copy the token and set it as a Collection Variable.
3. **Header Setup:** In the Collection "Auth" tab, select **Bearer Token** and use {{token}}.
4. **Create Test:** Send the POST / request first to ensure the database accepts your schema.
5. **Validation Test:** Try to send a dispatch_date that is *after* the expected_delivery_date.
    - **Expected Result:** 400 Bad Request with message "Dispatch Date cannot be after Expected Delivery Date".
6. **CSV Test:** Create a .csv file with the required headers and use the form-data body type to upload.

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5. Error Code Reference

Code	Type	Description
400	Bad Request	Validation error (Zod) or Duplicate shipment_id.
401	Unauthorized	Token is missing or expired.
403	Forbidden	User role lacks permission for this action.
404	Not Found	The id provided does not exist in the database.
500	Server Error	Unexpected database or system error.